Image Analysis Report

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Executive Summary

The flange and valve assembly exhibit severe atmospheric corrosion characterized by significant material loss on fasteners and heavy lamellar rust scaling. Based on the visual evidence, the corrosion has reached a critical stage where structural integrity of the bolting and sealing surfaces is likely compromised.

Key Findings

#	Finding
1	Severe lamellar (layered) iron oxide scale across the entire flange assembly and valve body.
2	Significant section loss on bolt heads and nuts, indicating they are likely seized and no longer provide design tension.
3	Corrosion product bridging across the flange gap, obscuring the gasket seating area and potential leak paths.
4	Complete coating failure on the flanges and valve body with 100% surface oxidation.
5	Localized pitting and coating delamination on the adjacent gray-painted pipe segment.
6	The internal condition of the valve, gasket, and flange faces cannot be determined from this external image.
7	Potential for microbial-induced corrosion (MIC) or crevice corrosion at the interface with the gravel substrate.

Recommendations

#	Recommendation
1	Perform Ultrasonic Thickness (UT) gauging on the pipe and valve body to verify remaining wall thickness meets the minimum required (T-min).
2	Schedule an immediate flange breakout to replace all fasteners with new B7/2H hardware or equivalent specifications.
3	Mechanical cleaning of all surfaces to SSPC-SP3 (Power Tool Cleaning) or SSPC-SP10 (Near-White Blast Cleaning) to assess the depth of pitting.
4	Conduct a Magnetic Particle Inspection (MPI) on the flange necks and valve body to check for surface-breaking cracks.
5	Replace the flange gasket after inspecting seating surfaces for wire-drawing or localized erosion.
6	Apply a high-performance epoxy coating system or cold-galvanizing compound to prevent further atmospheric attack.
7	Install flange protectors or use petrolatum-based anti-corrosion tape (e.g., Denso tape) for long-term protection in this high-moisture environment.
8	Perform a leak test or hydrotest at 1.1x operating pressure following the component replacements and reassembly.